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Pascal's Hyper-Pyramids

CPU TIME LIMIT
1 second

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1024 MB

DIFFICULTY
4.2 Medium

of numbers with 1 at the top and whose entries are the sum

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Pascal's triangle of height 5:

$$
\begin{array}{|lllll}
\hline
1 & 1 & 1 & 1 & \mathbf{1} \\
1 & 2 & 3 & \mathbf{4} & \\
1 & 3 & \mathbf{6} & & \\
1 & \mathbf{4} & & & \\
\mathbf{1} & & & &
\end{array}
$$

Pascal's hyper-pyramids generalize the triangle to higher dimensions. In 3 dimensions, the value at position (x, y, z) is the sum of up to three other values:

- $(x, y, z-1)$, the value immediately below it if we are not on the bottom face $(z = 0)$;
- $(x, y-1, z)$, the value immediately behind if we are not on the back face $(y = 0)$;
- $(x-1, y, z)$, the value immediately to the left if we are not on the leftmost face $(x = 0)$.

All values at the bottom, back, and leftmost faces are 1.

The following figure depicts Pascal's 3D-pyramid of height 5 as a series of plane cuts obtained by fixing the value of the z coordinate.

$z = 0$

1	1	1	1	**1**
1	2	3	**4**	
1	3	**6**		
1	**4**			
1				

$z = 1$

1	2	3	**4**
2	6	**12**	
3	**12**		
4			

$z = 2$

1	3	**6**
3	**12**	
6		

$z = 3$

1	**4**
4	

$z = 4$

1

For example, the number at position $x = 1$, $y = 2$, $z = 1$ is the sum of the values at $(0, 2, 1)$, $(1, 1, 1)$ and $(1, 2, 0)$, namely, $6 + 3 + 3 = 12$. The base of the pyramid corresponds to a plane of positions such that $x + y + z = 4$ (highlighted in bold above).

The size of each layer grows quadratically with the height of the pyramid, but there are many repeated values due to symmetries: numbers at positions that are permutations of one another must be equal. For example, the numbers at positions $(0, 1, 2)$, $(1, 2, 0)$ and $(2, 1, 0)$ above are all equal to 3.

Task

Write a program that, given the number of dimensions D of the hyper-space and the height H of a hyper-pyramid, computes the set of numbers at the base.

Input

A single line with two positive integers: the number of dimensions, D, and the height of the hyper-pyramid, H.

Constraints

2	$\leq$	D	$<$	32	Number of dimensions
1	$\leq$	H	$<$	32	Height

D and H are such that all numbers in the hyper-pyramid are less than 2^{63}.

Output

The set of numbers at the base of the hyper-pyramid, with no repetitions, one number per line, and in ascending order.

Sample Input 1

```
2 5
```

Sample Output 1

```
1
4
6
```

Sample Input 2

```
3 5
```

Sample Output 2

```
1
4
6
12
```